PROBLEM DA11

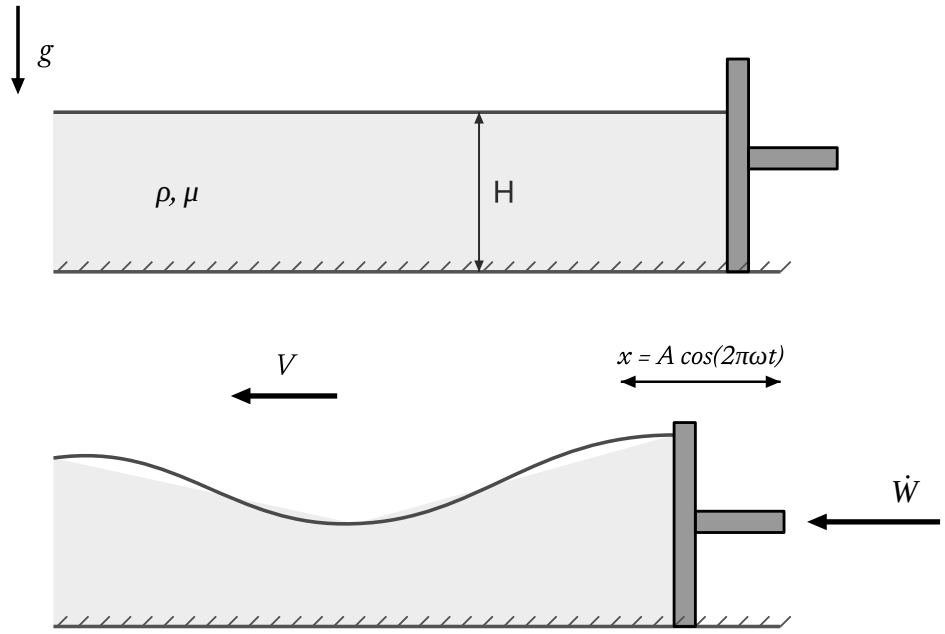
Power of a Wave Generator

Original statement · DA11.pdf

Π theorem | Froude similarity | Re ≫ 1

OFFICIAL PROBLEM STATEMENT

PROBLEM 3 (45 minutes) The figure below illustrates a simple mechanism to generate periodic gravity waves in a pool of liquid (density ρ and viscosity μ), with undisturbed height H. The waves are driven by the harmonic motion of the lateral confining wall perpendicular to itself, with amplitude A and frequency ω.



Statement figure (redrawn from DA11.pdf): above, the pool of liquid (ρ, μ) of undisturbed depth H; below, the lateral wall moved harmonically, x = A cos(2πωt), by the power Ẇ, generating gravity waves

1. (*4 points*). Use dimensional analysis to simplify the dependence of the power needed to drive the generator, $\dot{W}$, as a function of the governing parameters $(\rho, \mu, A, \omega, H, g)$.

2. (*1 point*). Simplify the resulting dimensionless function for the case of negligible viscous forces. In the remainder of the problem, assume that this condition is satisfied.

To characterize the requirements of a new wave generator to be installed in a large water channel (subindex p), a scale model (subindex m) with $H_m/H_p = 1/20$ is tested in the lab using water as working fluid. Knowing that the wave frequency required in the water channel is $\omega_p = 1/6\ \mathrm{s}^{-1}$,

3. (*2 points*). Determine the value of the frequency, ω_m, that must be used in the experiments to guarantee physical similarity.

4. (*2 points*). If the power used in the model is $\dot{W}_m = 1$ Watt, obtain the power needed to drive the wave generator in the water channel, $\dot{W}_p$.

5. (*1 point*). Justify the hypothesis of negligible viscous forces. If needed, make a reasonable guess for the values of the involved variables.

`Source: DA11.pdf (page 1: statement; page 2: handwritten key). Notes.pdf, Chapter 6: Π theorem, Eq. 6.25; limits, Eq. 6.55; physical similarity, Eqs. 6.57-6.59.`

Worked solution

Hypotheses, Data and Dimensional Table

- Gravity waves at a free surface: g is essential (Notes.pdf, p. 66). The wall motion $x = A\cos(2\pi\omega t)$ shows that ω is a frequency in cycles per second, $[\omega] = T^{-1}$.
- The model is geometrically similar (all lengths, the channel width included, scaled by $1/20$) and uses water, as the channel: same ρ, μ, g.

Parameter	Dimensions
$\dot{W}$ (unknown)	ML^2T^{-3}
ρ	ML^{-3}
μ	$ML^{-1}T^{-1}$
A, H	L
ω	T^{-1}
g	LT^{-2}

Application of the Π Theorem

$$\dot{W} = f(\rho, \mu, A, \omega, H, g)$$

$n = 6$. The key takes (ρ, A, ω): rows $\rho(-3, 1, 0)$, $A(1, 0, 0)$, $\omega(0, 0, -1)$ in (L, M, T); expanding along the ω row, $\det = (-1) \cdot ((-3) \cdot 0 - 1 \cdot 1) = 1 \neq 0$, so $k = 3$ and three groups remain.

Power. $[\dot{W}] = [\rho]^a [\omega]^b [A]^c$: $M:\ 1 = a$; $T:\ -3 = -b$; $L:\ 2 = -3 + c$; so $\Pi_{\dot{W}} = \dot{W}/(\rho \omega^3 A^5)$.

Viscosity. $[\mu] = [\rho]^a [\omega]^b [A]^c$: $M:\ 1 = a$; $T:\ -1 = -b$; $L:\ -1 = -3 + c$; so $\Pi_\mu = \mu/(\rho \omega A^2)$, the inverse of a Reynolds number with the wall velocity ωA.

Depth. $\Pi_H = H/A$.

Gravity. $[g] = [\omega]^b [A]^c$: $T:\ -2 = -b$; $L:\ 1 = c$; so $\Pi_g = g/(\omega^2 A)$.

$$\Pi_{\dot{W}} = \Pi_{\dot{W}}(\Pi_\mu, \Pi_H, \Pi_g): \qquad \frac{\dot{W}}{\rho \omega^3 A^5} = \Pi_{\dot{W}}\left(\frac{\mu}{\rho \omega A^2}, \frac{H}{A}, \frac{g}{\omega^2 A}\right)$$

Negligible Viscous Forces

For $\Pi_\mu \to 0$ the group drops out (Notes.pdf, Eq. 6.55):

$$\frac{\dot{W}}{\rho \omega^3 A^5} = \Pi_{\dot{W}}\left(\frac{H}{A}, \frac{g}{\omega^2 A}\right)$$

Physical Similarity

Depth group. Geometric similarity of the whole setup, amplitude included:

$$\frac{H_p}{A_p} = \frac{H_m}{A_m} \implies \frac{A_p}{A_m} = \frac{H_p}{H_m} = 20$$

Gravity group. Same g:

$$\frac{g}{\omega_p^2 A_p} = \frac{g}{\omega_m^2 A_m} \implies \frac{\omega_m}{\omega_p} = \sqrt{\frac{A_p}{A_m}} = \sqrt{20}$$

$$\omega_m = \sqrt{20} \times \frac{1}{6} = \frac{2\sqrt{5}}{6} = \frac{\sqrt{5}}{3}\ \mathrm{s}^{-1} = 0.745\ \mathrm{s}^{-1}$$

Power of the Prototype

With Π_H and Π_g equal, $\Pi_{\dot{W}}$ is equal (same ρ):

$$\frac{\dot{W}_p}{\rho \omega_p^3 A_p^5} = \frac{\dot{W}_m}{\rho \omega_m^3 A_m^5} \implies \frac{\dot{W}_p}{\dot{W}_m} = \left(\frac{\omega_p}{\omega_m}\right)^3 \left(\frac{A_p}{A_m}\right)^5 = \left(\frac{A_p}{A_m}\right)^{-3/2} \left(\frac{A_p}{A_m}\right)^5 = \left(\frac{A_p}{A_m}\right)^{7/2}$$

$$\dot{W}_p = 20^{7/2} \times 1\ \mathrm{W} = 20^3 \times \sqrt{20}\ \mathrm{W} = 8000 \times 4.472 = 35\,777\ \mathrm{W} = 35.78\ \mathrm{kW}$$

Negligible Viscosity: Justification

For water, $\rho = 10^3$ kg/m^3 and $\mu = 10^{-3}$ kg/(m s), so $\Pi_\mu = 10^{-6}/(\omega A^2)$ with ω in s^{-1} and A in m. The amplitudes are not given; reasonable guesses, as the key suggests:

- channel: $A_p \sim 1$ m, $\omega_p = 1/6$ s^{-1}, so $\Pi_\mu \sim 10^{-6}/(0.17 \times 1) = 6 \times 10^{-6} \ll 1$;
- model: $A_m = A_p/20 \sim 0.05$ m, $\omega_m = 0.745$ s^{-1}, so $\Pi_\mu \sim 10^{-6}/(0.745 \times 0.0025) = 5 \times 10^{-4} \ll 1$.

Viscous forces are negligible in both (large Reynolds numbers).

Answers and Checks

- Reduced law: $\dot{W}/(\rho\omega^3 A^5) = \Pi_{\dot{W}}\left(\mu/(\rho\omega A^2),\ H/A,\ g/(\omega^2 A)\right)$; without viscosity, $\Pi_{\dot{W}}(H/A, g/(\omega^2 A))$.
- Model frequency: $\omega_m = \sqrt{5}/3 = 0.745\ \text{s}^{-1}$.
- Channel power: $\dot{W}_p = 20^{7/2}\,\dot{W}_m = 35.78\ \text{kW}$.
- Dimensions: $\rho\omega^3 A^5$ is $ML^{-3}\cdot T^{-3}\cdot L^5 = ML^2T^{-3}$, a power.
- Froude scaling: times scale as $\lambda^{1/2}$ and powers as $\lambda^{7/2}$ with $\lambda = 20$, as for the ship of DA1.
- Check against the official key: DA11.pdf gives $\omega_m = \sqrt{5}/3\ \text{s}^{-1}$ and $\dot{W}_p = 20^{7/2}\dot{W}_m = 35.78$ kW, and justifies $\Pi_\mu \ll 1$ with $A \sim 1$ m, $\omega \sim 1\ \text{s}^{-1}$.